Experimental Deisgn Final

Joseph Ortega

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```r
library(tidyverse)
```

```r
# Dataset

reading_speed = read.csv("C:/Users/jorte/Downloads/Experiment 158 - R Studio Version.csv")
reading_speed
```

```
##    id Y_i.0...TNR. Y_i.1...CS. D_i
## 1   1         1:25        1:45   0
## 2   2         2:23        2:01   0
## 3   3         1:33        2:04   0
## 4   4         5:40        4:31   0
## 5   5         2:04        1:40   0
## 6   6         1:48        1:44   0
## 7   7         2:57        2:51   0
## 8   8         4:37        3:57   0
## 9   9         3:10        3:40   0
## 10 10         2:07        2:14   0
## 11 11         3:27        3:44   0
## 12 12         2:29        2:43   0
## 13 13         3:16        2:56   0
## 14 14         2:26        2:28   1
## 15 15         1:52        2:04   1
## 16 16         1:44        1:37   1
## 17 17         1:13        0:53   1
## 18 18         2:07        1:13   1
## 19 19         1:27        1:41   1
## 20 20         2:46        2:33   1
## 21 21         2:47        3:38   1
## 22 22         2:34        2:21   1
## 23 23         2:55        2:57   1
## 24 24         3:34        2:50   1
## 25 25         1:21        1:15   1
## 26 26         4:34        4:54   1
```

```r
# Convert time into seconds

reading_speed = reading_speed %>%
  mutate(
    Y0_sec = as.numeric(sub(":.*", "", Y_i.0...TNR.)) * 60 +
             as.numeric(sub(".*:", "", Y_i.0...TNR.)),

    Y1_sec = as.numeric(sub(":.*", "", Y_i.1...CS.)) * 60 +
```

```r
            as.numeric(sub(".*:", "", Y_i.1...CS.))
  )

reading_speed
```

```
##    id Y_i.0...TNR. Y_i.1...CS. D_i Y0_sec Y1_sec
## 1   1         1:25        1:45   0     85    105
## 2   2         2:23        2:01   0    143    121
## 3   3         1:33        2:04   0     93    124
## 4   4         5:40        4:31   0    340    271
## 5   5         2:04        1:40   0    124    100
## 6   6         1:48        1:44   0    108    104
## 7   7         2:57        2:51   0    177    171
## 8   8         4:37        3:57   0    277    237
## 9   9         3:10        3:40   0    190    220
## 10 10         2:07        2:14   0    127    134
## 11 11         3:27        3:44   0    207    224
## 12 12         2:29        2:43   0    149    163
## 13 13         3:16        2:56   0    196    176
## 14 14         2:26        2:28   1    146    148
## 15 15         1:52        2:04   1    112    124
## 16 16         1:44        1:37   1    104     97
## 17 17         1:13        0:53   1     73     53
## 18 18         2:07        1:13   1    127     73
## 19 19         1:27        1:41   1     87    101
## 20 20         2:46        2:33   1    166    153
## 21 21         2:47        3:38   1    167    218
## 22 22         2:34        2:21   1    154    141
## 23 23         2:55        2:57   1    175    177
## 24 24         3:34        2:50   1    214    170
## 25 25         1:21        1:15   1     81     75
## 26 26         4:34        4:54   1    274    294
```

```r
# Calculate ATE and t test

reading_speed = reading_speed %>%
  mutate(diff = Y1_sec - Y0_sec)

mean(reading_speed$diff)
```

```
## [1] -4.692308
```

```r
t.test(reading_speed$diff)
```

```
##
##  One Sample t-test
##
## data:  reading_speed$diff
## t = -0.86518, df = 25, p-value = 0.3952
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  -15.862276   6.477661
## sample estimates:
## mean of x
## -4.692308
```

```r
# Regression model

lm(diff ~ D_i, data = reading_speed)
```

```
##
## Call:
## lm(formula = diff ~ D_i, data = reading_speed)
##
## Coefficients:
## (Intercept)          D_i
##     -5.0769       0.7692
```

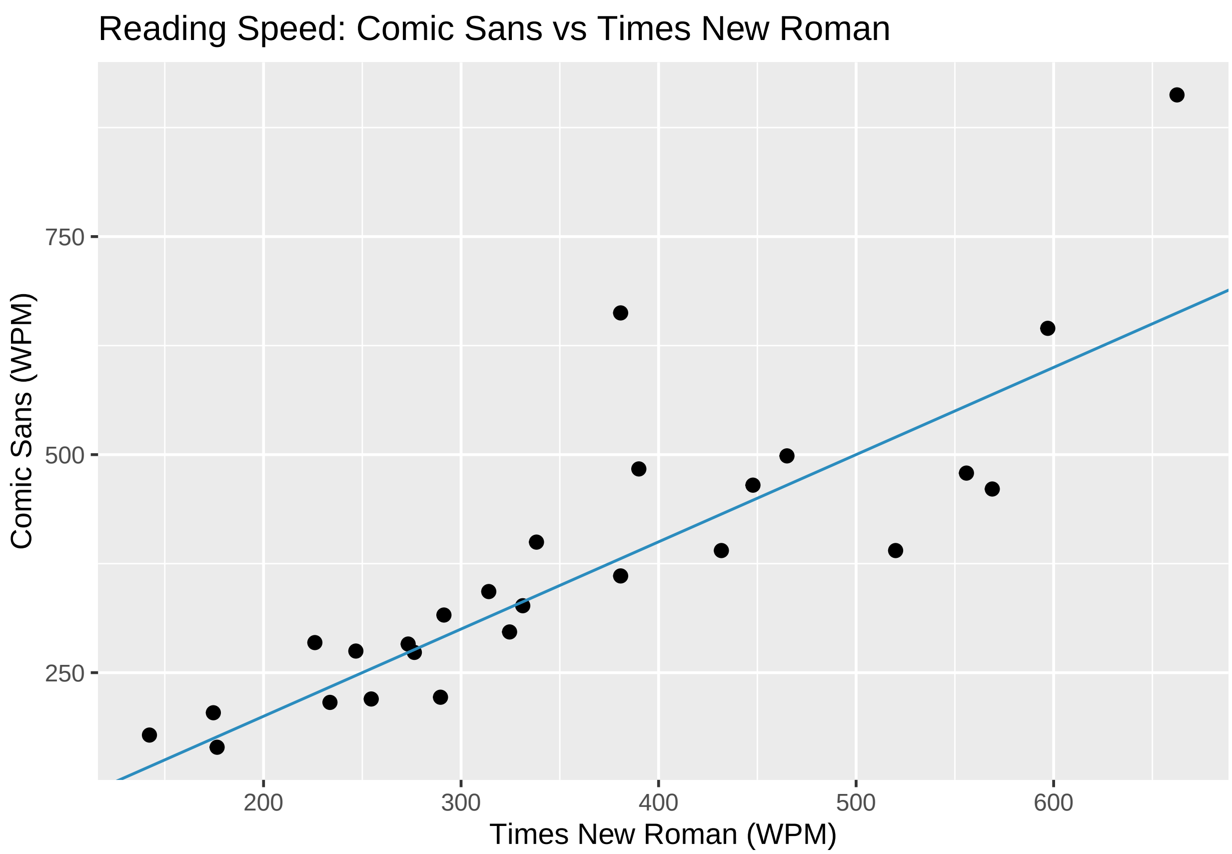
```r
# Create WPM columns

words = 806

reading_speed = reading_speed %>%
  mutate(
    wpm_TNR = words / (Y0_sec / 60),
    wpm_CS  = words / (Y1_sec / 60)
  )
```

```r
# Plot

ggplot(reading_speed, aes(x = wpm_TNR, y = wpm_CS)) +
  geom_point(size = 2) +
  geom_abline(slope = 1, intercept = 0, color = "#2b8cbe") +
  labs(
    title = "Reading Speed: Comic Sans vs Times New Roman",
    x = "Times New Roman (WPM)",
    y = "Comic Sans (WPM)"
  )
```




```r
# Random assignment 1000 simulations

set.seed(123)

B = 1000
ri_stats = numeric(B)

obs_stat = mean(reading_speed$diff)

for (b in 1:B) {
  flip = sample(c(-1, 1), size = nrow(reading_speed), replace = TRUE)
  ri_stats[b] = mean(reading_speed$diff * flip)
}

p_value = mean(abs(ri_stats) >= abs(obs_stat))

# Results
obs_stat
```

```
## [1] -4.692308
```

```r
p_value
```

```
## [1] 0.412
```

```r
ggplot(data.frame(ri_stats), aes(x = ri_stats)) +
  geom_histogram(bins = 30, fill = "#80cdc1", color = "#003c30") +
  geom_vline(xintercept = obs_stat, color = "#8c510a", linewidth = 1) +
```

```r
  labs(
    title = "Randomization Inference Null Distribution",
    x = "Simulated ATE (seconds)",
    y = "Frequency"
  )
```